

Kinematics and Dynamics of Machines MEL3020

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Velocity Analysis using instant centers

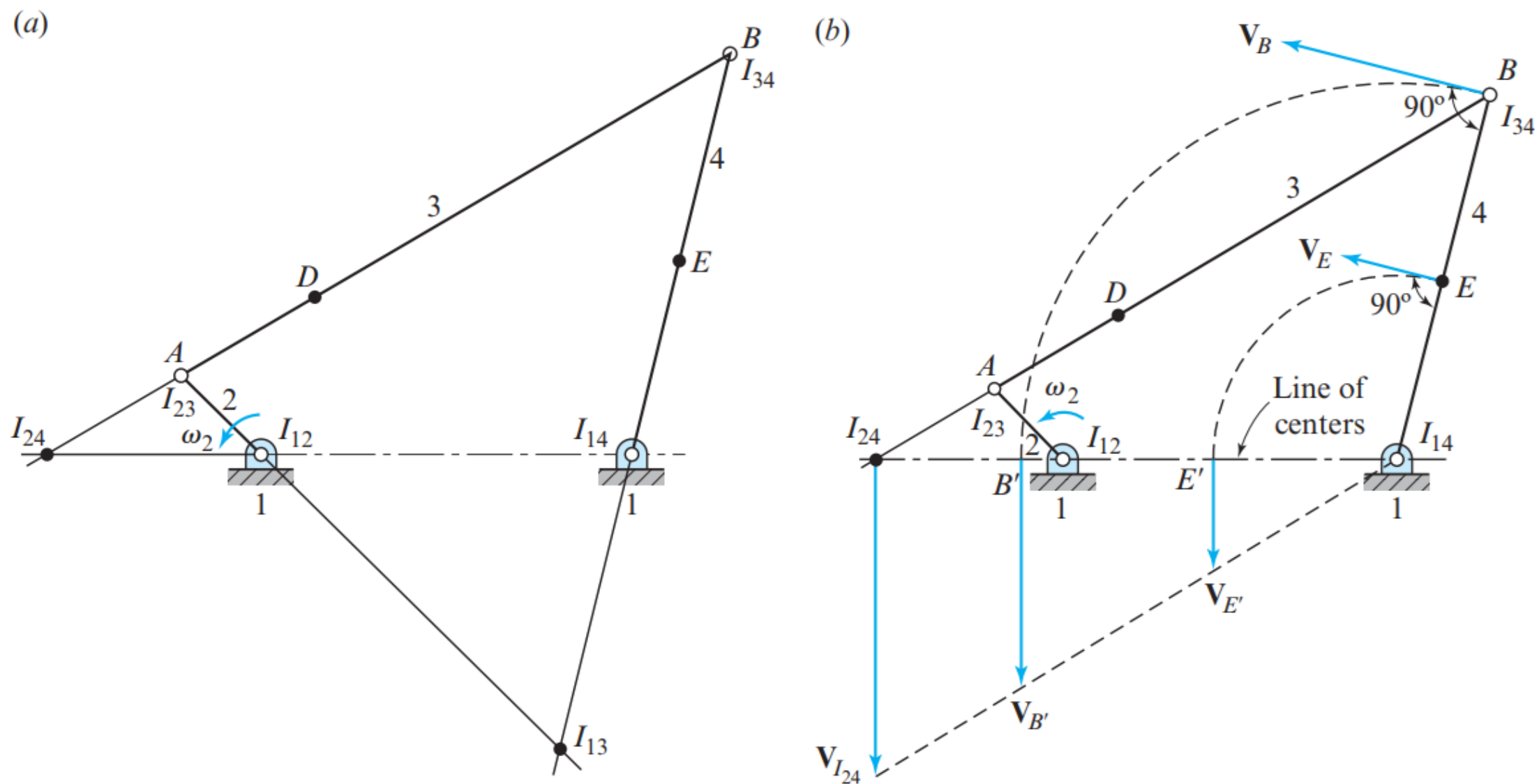


Figure 3.29 Graphic velocity analysis using instant centers.

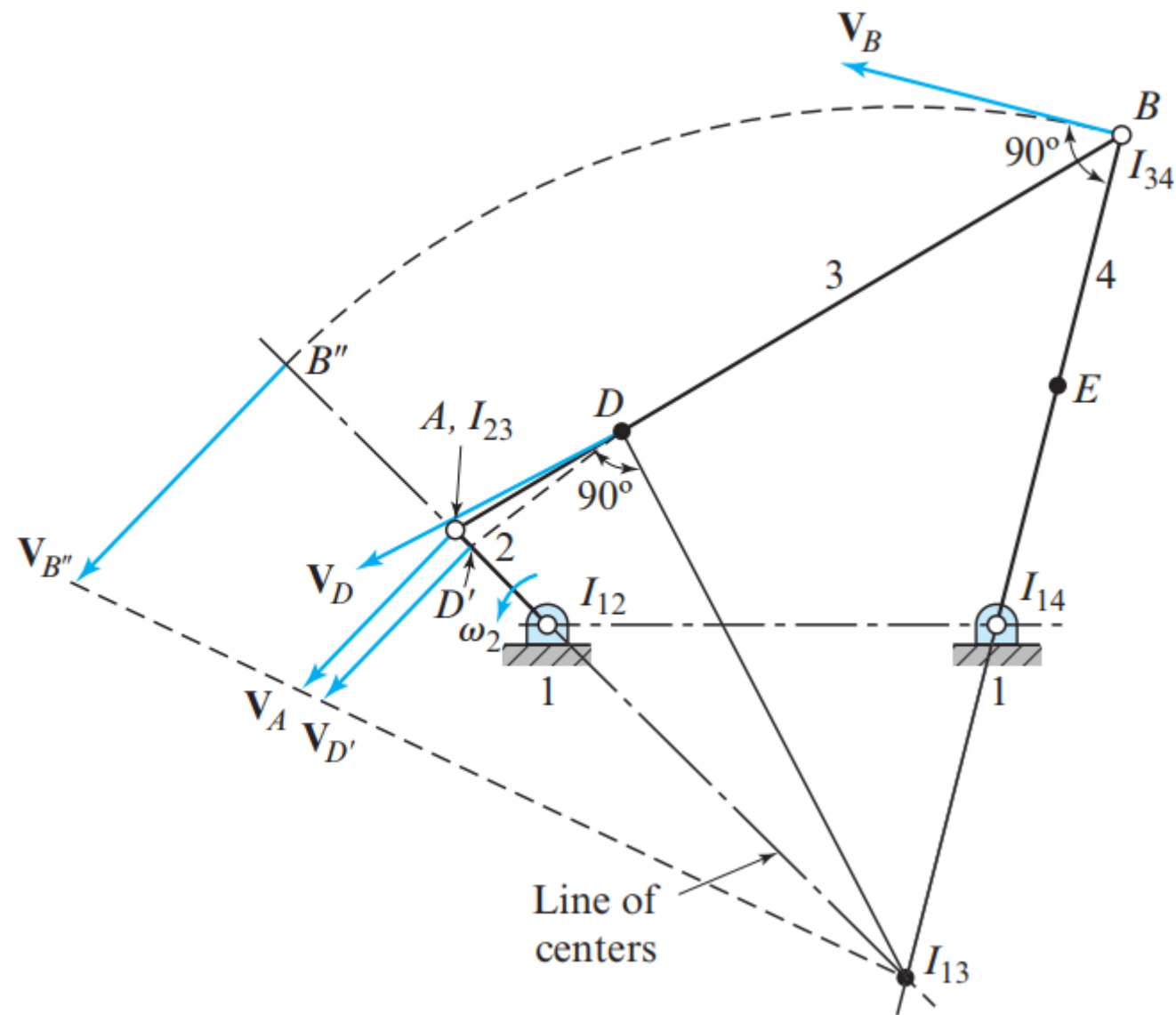


Figure 3.30 Velocities of pin B and point D .

EXAMPLE 3.12

For the device shown in Fig. 3.31, some of the links are visible, whereas other links are enclosed in a housing. However, the location of instant center I_{25} is known, as shown in the figure. If the velocity of point C of link 6 is $\mathbf{V}_C = 10$ m/s to the right, determine the angular velocity of link 2.

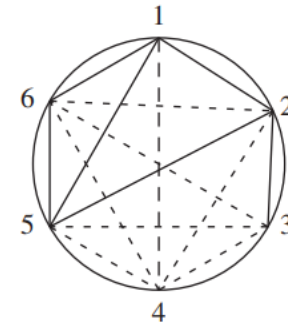
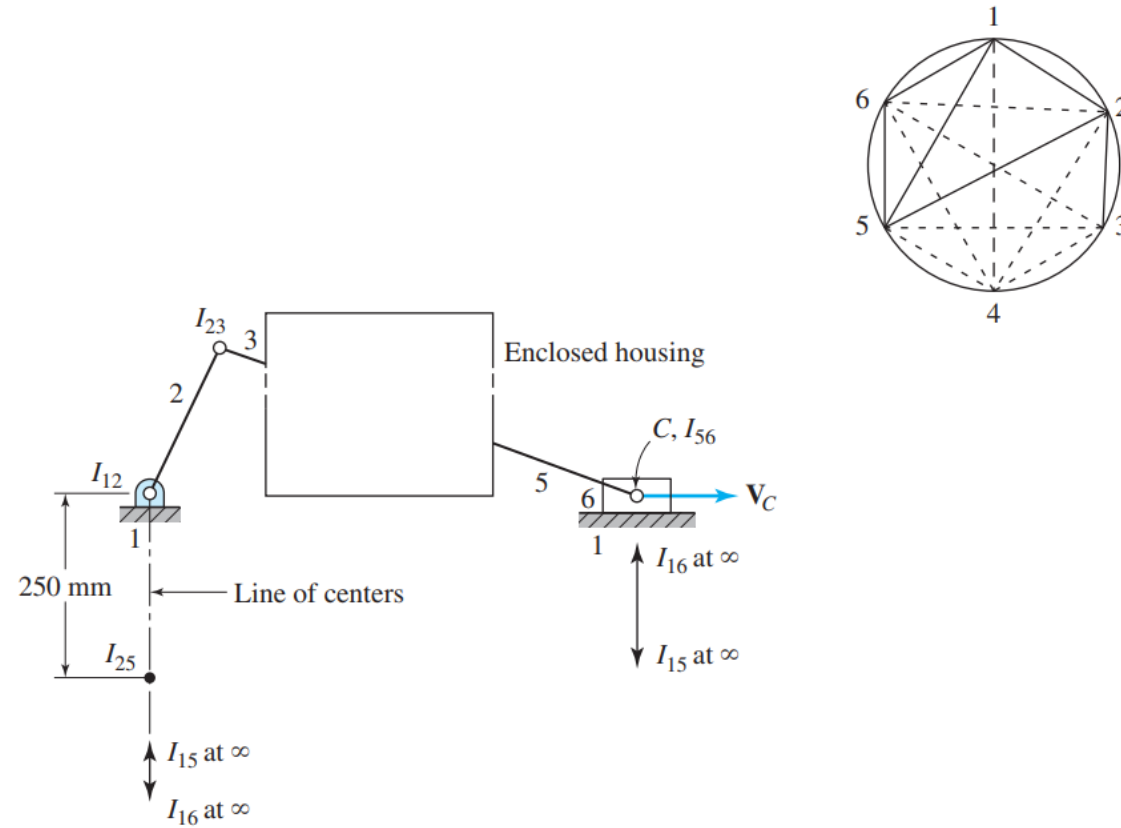
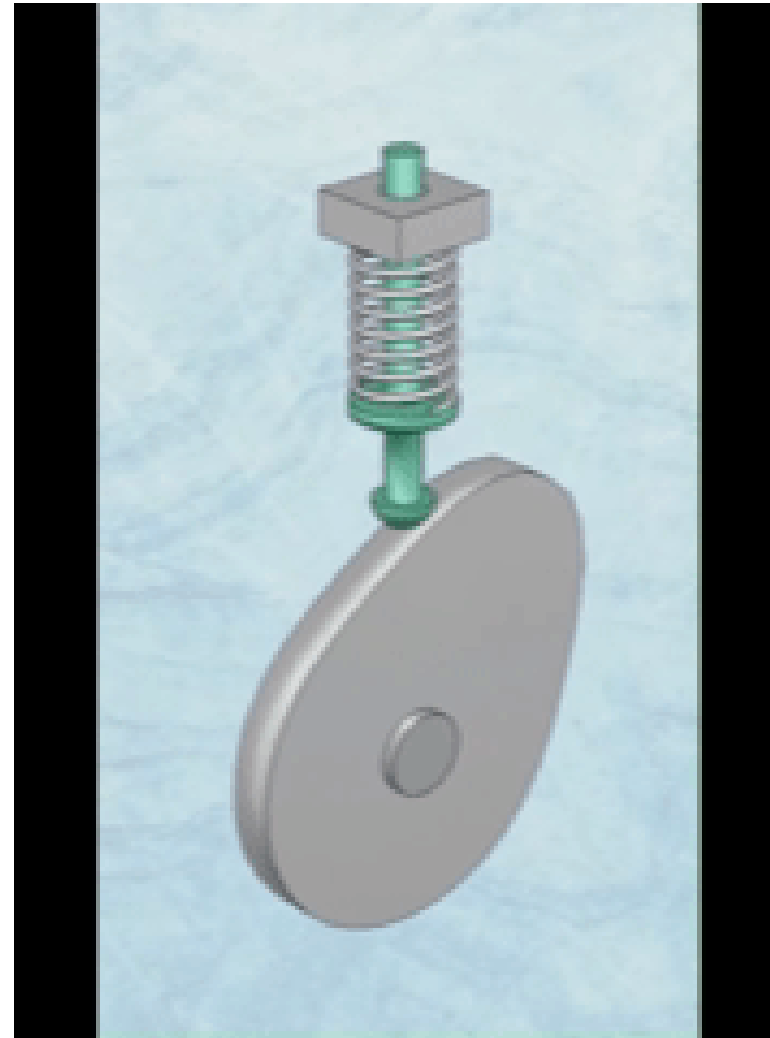
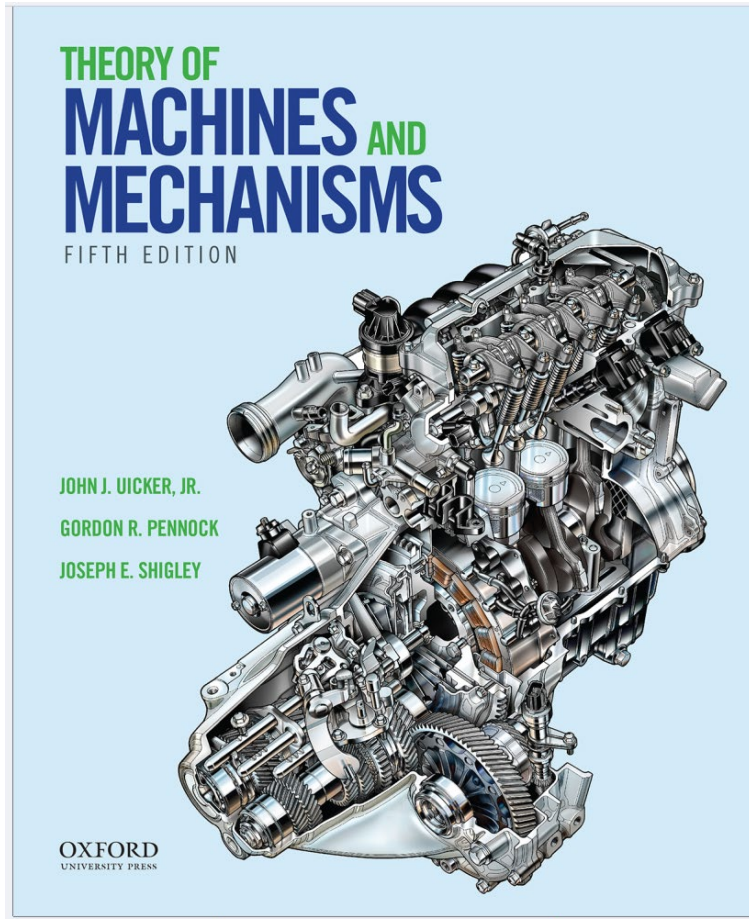


Figure 3.31 Links enclosed in a housing.

CAM Analysis



CAM Classification

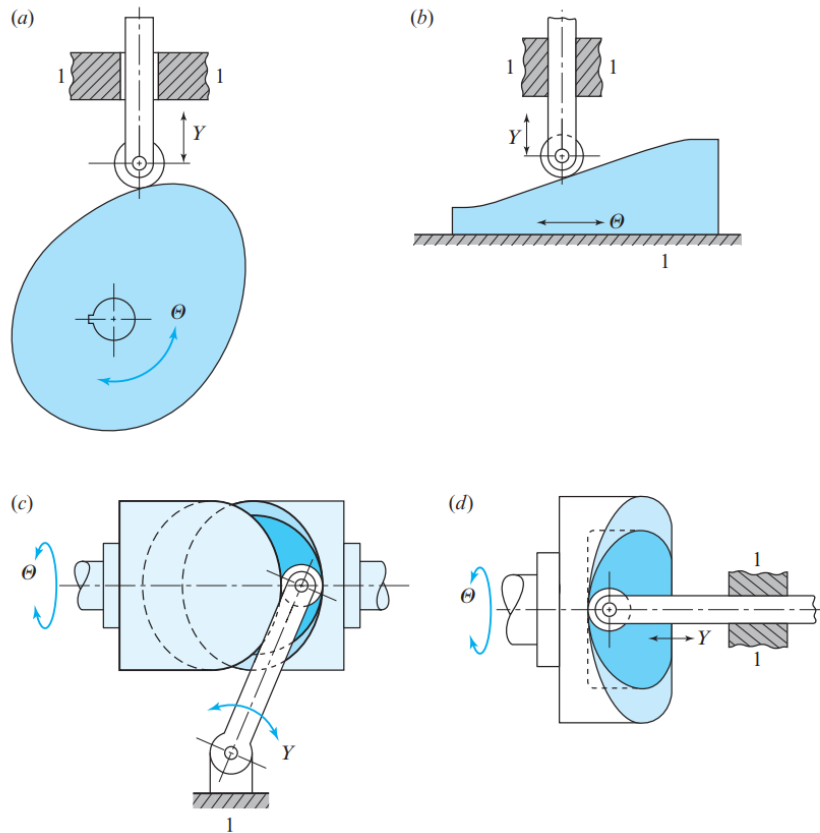


Figure 6.1 (a) Plate cam; (b) wedge cam; (c) barrel cam; (d) face cam.

CAM Classification

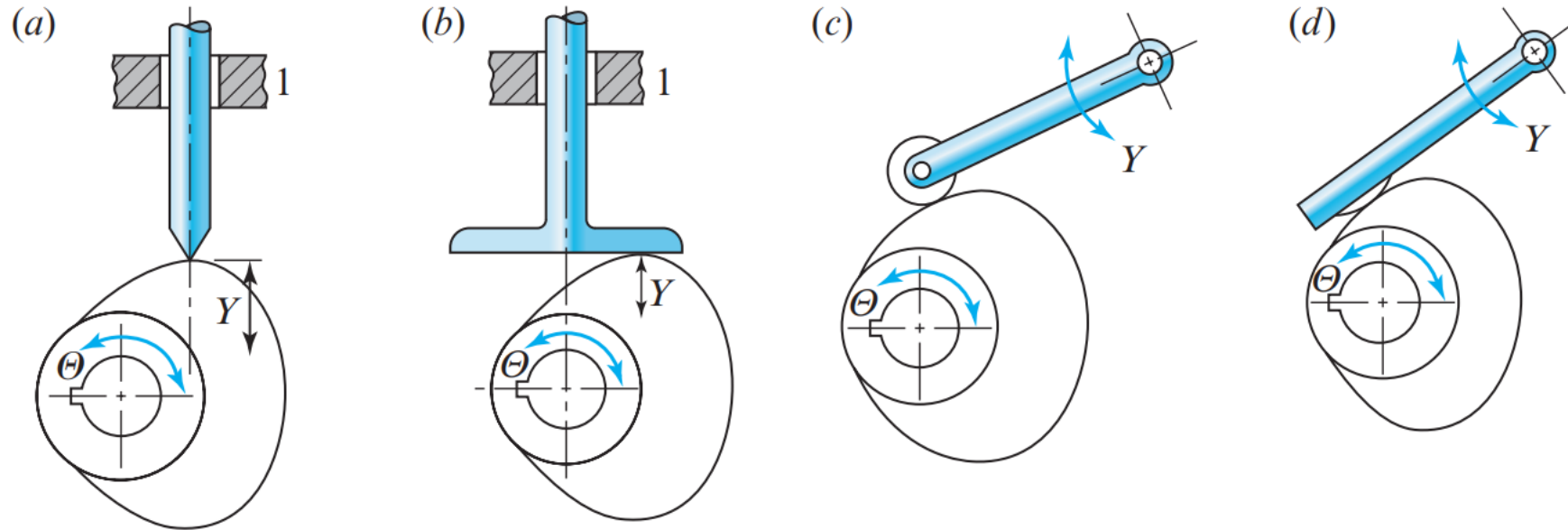


Figure 6.2 Plate cams with (a) an offset reciprocating knife-edge follower; (b) a reciprocating flat-face follower; (c) an oscillating roller follower; (d) an oscillating curved-shoe follower.

CAM Classification

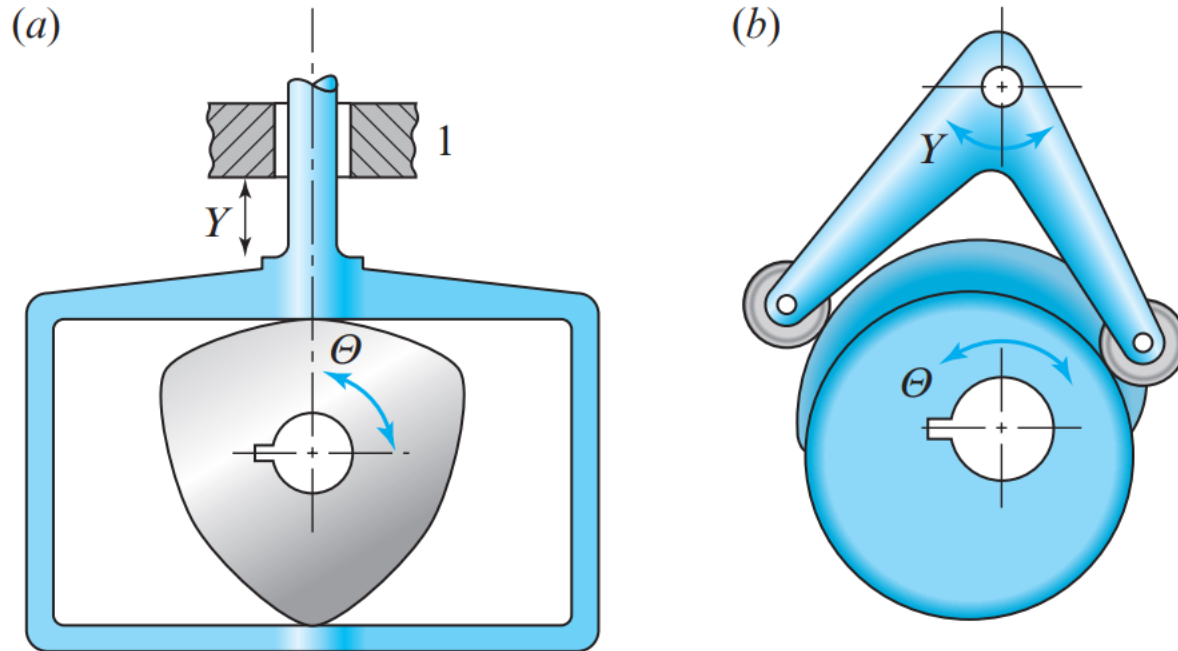


Figure 6.3

(a) Constant-breadth cam with a reciprocating flat-face follower; (b) conjugate cams with an oscillating roller follower.

Displacement Diagrams

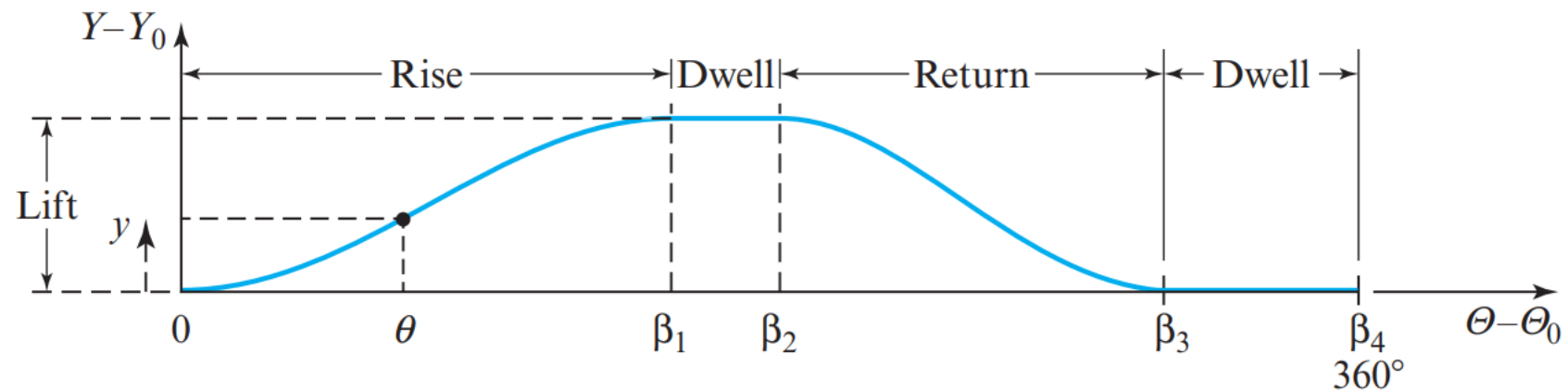


Figure 6.4 Displacement diagram for a cam.

Example

Cam Profile Coordinates for Translating Roller Follower

Determine the cam profile assuming that the follower starts with a dwell from 0° to 90° and rotates CW. The rise occurs with cycloidal motion during the cam rotation from 90° to 180° . The follower then dwells for 60° of cam rotation, and the return occurs with harmonic motion for the cam rotation from 240° to 360° . The amplitude of the follower translation is 2 cm, and the follower radius is 1 cm. The base circle radius is 4 cm, and the offset is 0.5 cm.

Example

